

# Minimum Energy Principle for Dynamic Cyclic Causality

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## Minimum Energy Principle

Resting-state fMRI networks exhibit strong recurrent feedback that is difficult to characterize using acyclic or pairwise causal models alone. We address this limitation by explicitly constructing a Hilbert space of cyclic interactions.

Directed interactions are represented as an edge-flow vector  $X(t) \in \mathbb{R}^{|E|}$ . The associated boundary matrices  $\mathbf{B}_1$  and  $\mathbf{B}_2$  encode node-edge and edge-face incidence, and define the 1-Hodge Laplacian (Anand and Chung, 2023)

$$\Delta_1 = \mathbf{B}_1^\top \mathbf{B}_1 + \mathbf{B}_2 \mathbf{B}_2^\top.$$

Under dissipative Lagrangian dynamics, the edge flow  $X(t)$  evolves toward configurations that minimize the Dirichlet potential energy (Chung et al., 2026a,b)  $E(X) = \frac{1}{2} X^\top \Delta_1 X$ . The resulting gradient flow is the Dirichlet-Hodge diffusion

$$\frac{dX(t)}{dt} = -\nabla E(X(t)) = -\Delta_1 X(t).$$

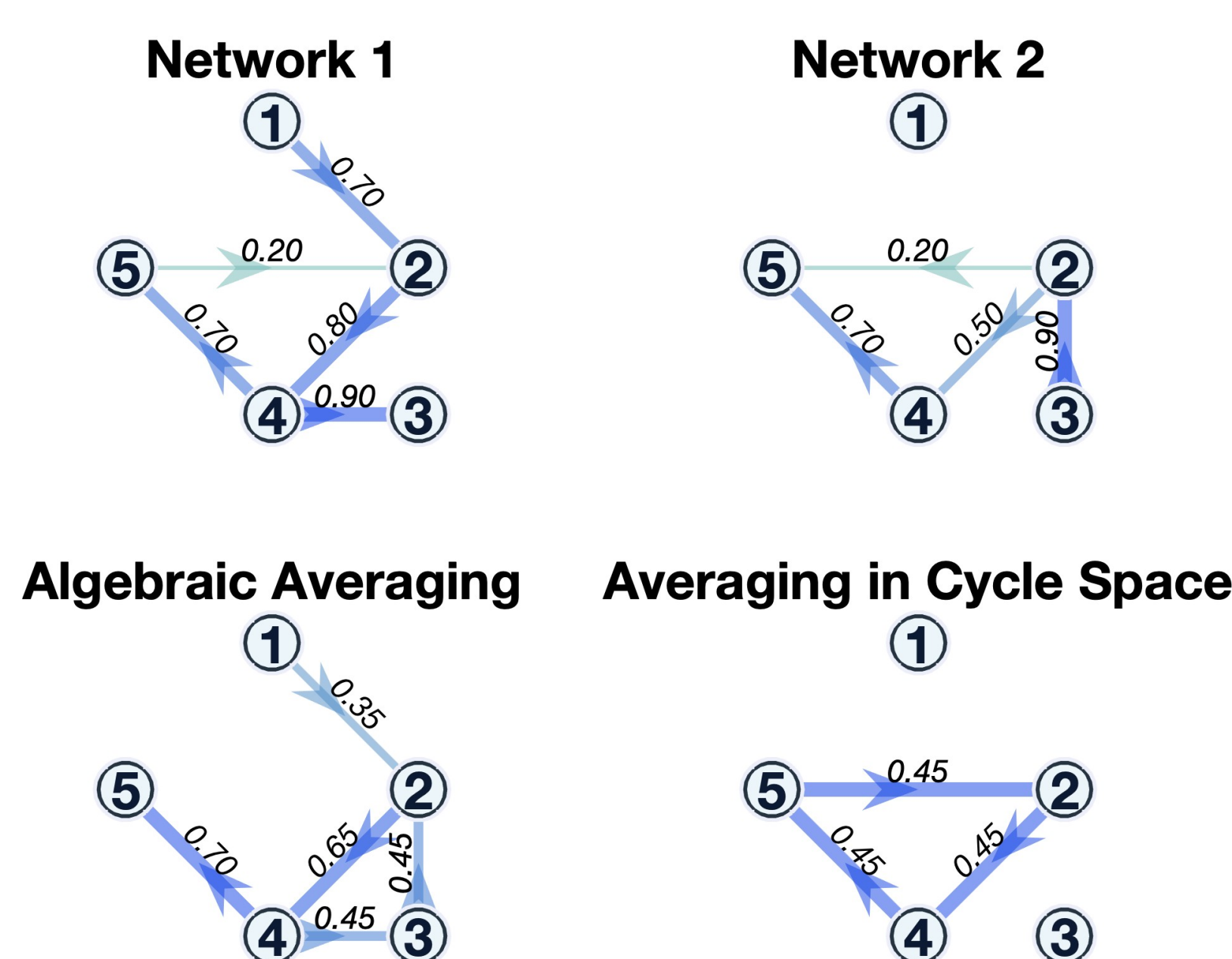
Along this trajectory, the energy decreases monotonically toward the zero-energy state

$$\ker \Delta_1 = \{X \in \mathbb{R}^{|E|} : \Delta_1 X = 0\}.$$

## Hilbert Space of Cycles

The cycle space  $\ker \Delta_1$  consists of cyclic interactions (Anand and Chung, 2023). Its dimension  $\beta_1 = \dim \ker(\Delta_1)$  is the first Betti number, giving the number of independent cycles. Let  $\{\phi_1, \dots, \phi_{\beta_1}\}$  denote the orthonormal eigenvectors of  $\Delta_1$  corresponding to zero eigenvalues. The cyclic component  $X_H$  of an edge flow  $X$  is obtained by orthogonal projection onto  $\ker \Delta_1$ :

$$X_H = \mathcal{P}_H X, \quad \mathcal{P}_H = \sum_{k=1}^{\beta_1} \phi_k \phi_k^\top.$$



**Fig. 1.** Two networks contain partially inconsistent cyclic structure involving  $2 \rightarrow 4 \rightarrow 5$ . Direct algebraic averaging removes this cycle and instead introduces a spurious cycle  $2 \rightarrow 4 \rightarrow 3$  that is not present in either network. In contrast, averaging in the harmonic cycle space enhances the coherent recurrent cycle  $5 \rightarrow 2 \rightarrow 4 \rightarrow 5$ .

Since  $\ker \Delta_1$  is a Hilbert space, averaging of cycles are well-defined operations that preserve cyclic consistency (Fig. 1). The inner product  $\langle X_H, Y_H \rangle = X_H^\top Y_H$  measures cyclic alignment and similarity between recurrent flow patterns.

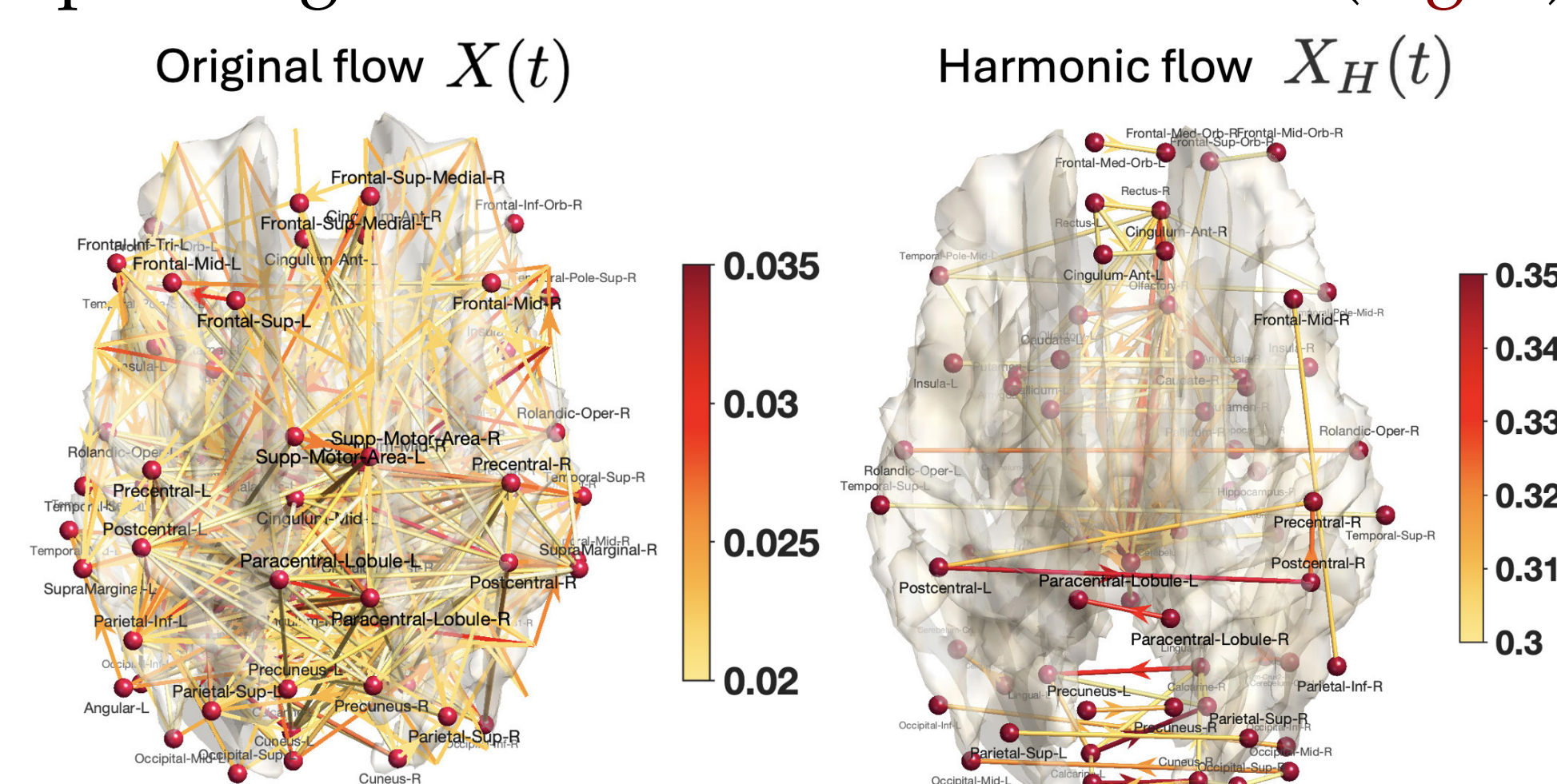
Cyclic interactions  $X_H$  have substantially lower variability than  $X$ . The total variance after harmonic projection satisfies

$$\frac{\sum_{i=|E|-\beta_1+1}^{|E|} \sigma_i}{\sum_{i=1}^{|E|} \sigma_i} \leq \frac{\text{Var}(X_H)}{\text{Var}(X)} = \frac{\text{tr}(\mathcal{P}_H \Sigma_X)}{\text{tr}(\Sigma_X)} \leq \frac{\sum_{i=1}^{\beta_1} \sigma_i}{\sum_{i=1}^{|E|} \sigma_i},$$

where  $\sigma_1 \geq \dots \geq \sigma_{|E|} \geq 0$  are the eigenvalues of  $\Sigma_X$ . Under isotropic covariance, this reduces to

$$\frac{\text{Var}(X_H)}{\text{Var}(X)} = \frac{\beta_1}{|E|}.$$

When  $\beta_1 \ll |E|$ ,  $\mathcal{P}_H$  contracts edge-flow variability into a low-dimensional cycle space. In our study,  $\beta_1/|E| = 534/6670 \approx 0.08$ , corresponding to 12-fold variance reduction (Fig. 2).



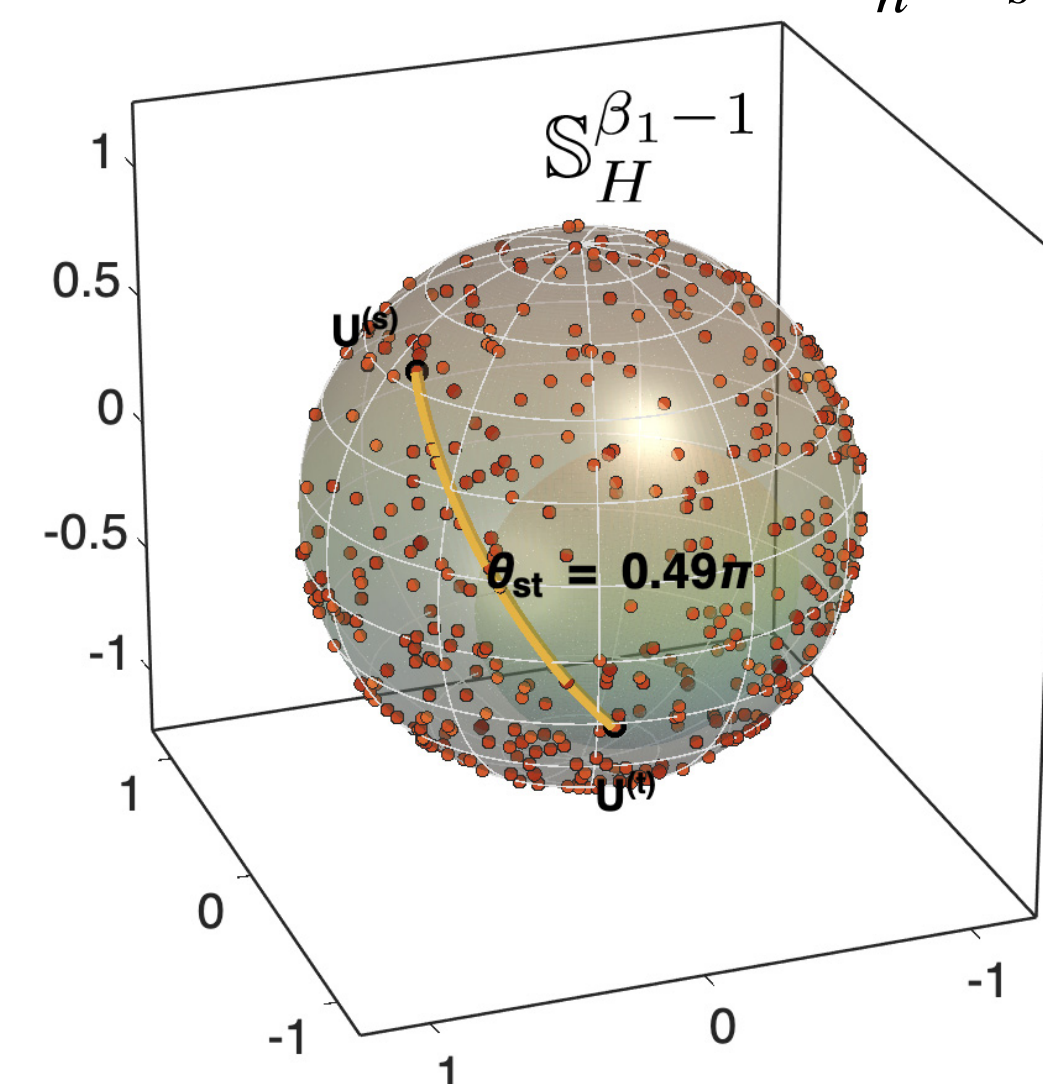
**Fig. 2.** Naive averaging of lagged correlation edge flows across time and 400 subjects yields weak diffuse connectivity (left; max 0.034). Harmonic averaging reduces transient edgewise fluctuations, increases the signal intensity, and recovers recurrent visual and sensorimotor organization (right).

## Population Inference in the Cycle Space

Let  $X_H^{(s)} \in \ker(\Delta_1)$  denote the harmonic flow for subject  $s$ . To remove variation in interaction magnitude across individuals, we normalize each flow to the harmonic unit sphere (Fig. 3):

$$U^{(s)} = \frac{X_H^{(s)}}{\|X_H^{(s)}\|_2} \in \mathbb{S}_H^{\beta_1-1}.$$

Population-level coherence is measured by the mean direction  $\bar{U} = \frac{1}{n} \sum_{s=1}^n U^{(s)}$ .



**Fig. 3.** Normalized harmonic flows from 400 subjects lie on  $\mathbb{S}_H^{\beta_1-1}$  and are embedded onto  $\mathbb{S}^2$  using MDS for visualization. The arc shows the geodesic angle  $\cos \theta_{st} = \langle U^{(s)}, U^{(t)} \rangle$ .

Under the null hypothesis of no preferred orientation on  $\mathbb{S}_H^{\beta_1-1}$ ,  $\|\bar{U}\|_2$  approaches zero, whereas shared cyclic organization produces a significantly larger mean direction (Fig. 4). We evaluate this coherence using the Rayleigh test (Mardia and Jupp, 2009),

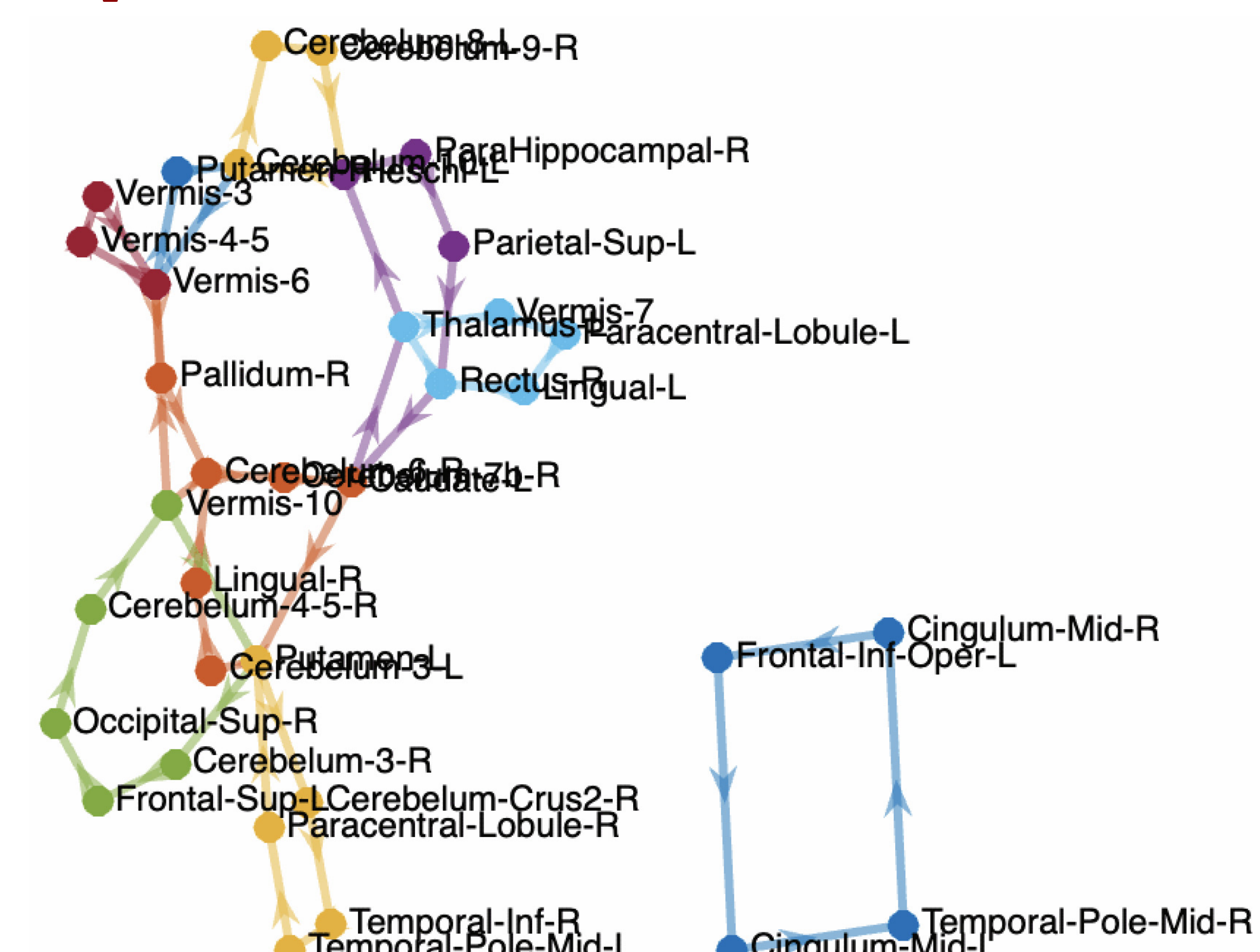
$$\beta_1 n \|\bar{U}\|_2^2 \xrightarrow{d} \chi_{\beta_1}^2,$$

from which the  $p$ -value is determined.

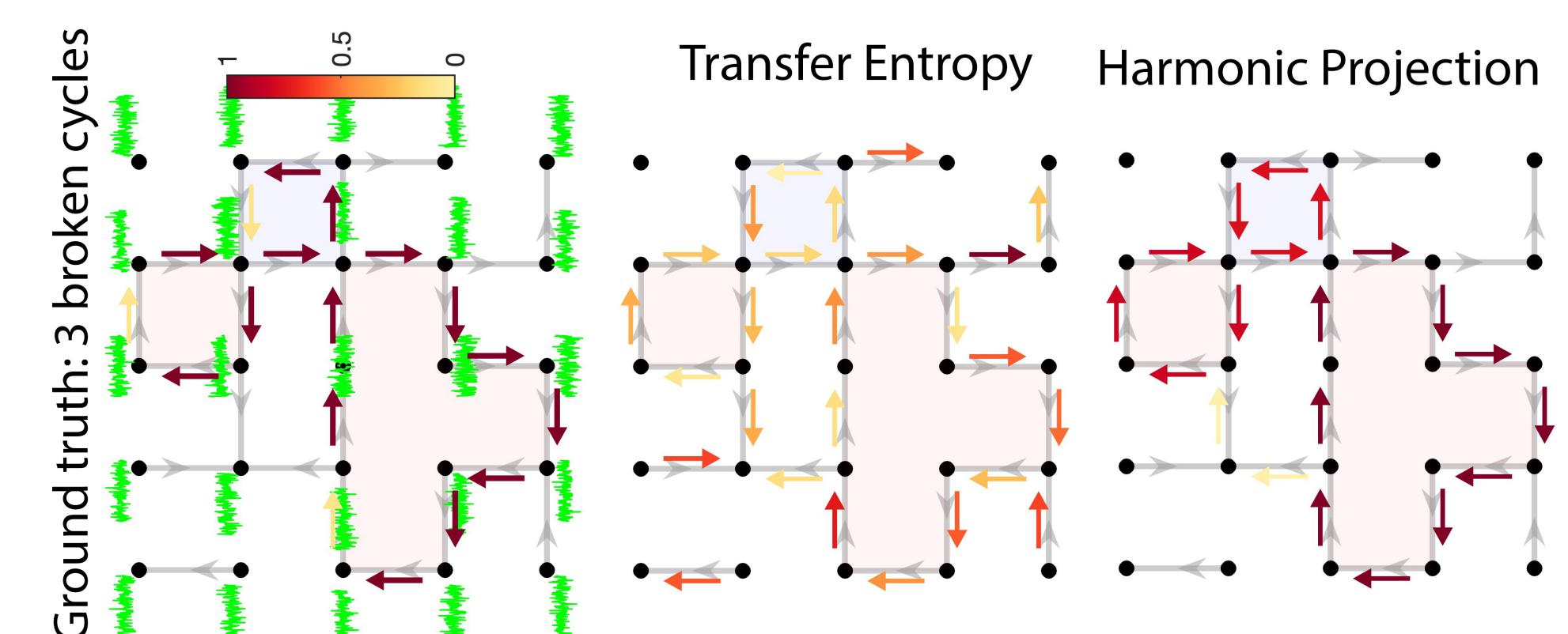
## Extensive Validation

We validated the method using synthetic dynamic networks with known directional

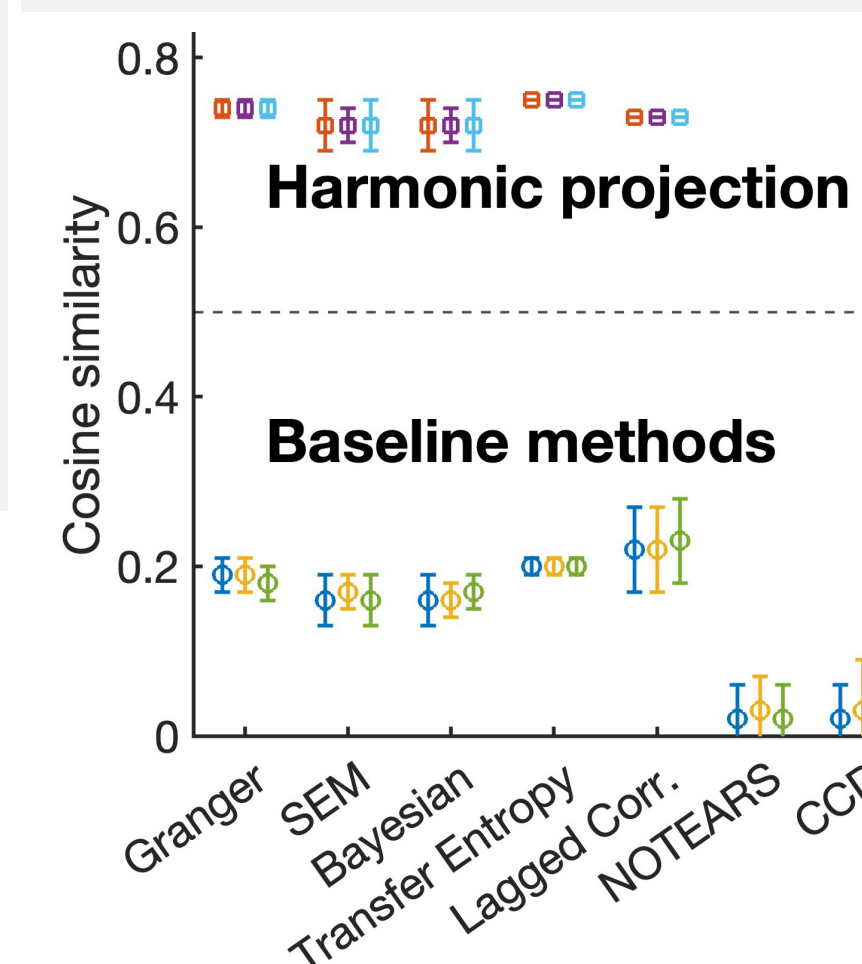
ground truth. Fig. 5 shows a stress test with three interacting cycles, where one edge per cycle is weakened from 1 to 0.1, below the noise level, masking cyclic structure in the observations. Against seven baselines (Fig. 6), harmonic projection improved cosine recovery from approximately 0.16–0.23 to 0.72–0.75 across noise levels  $\sigma = 0.1, 1, 10$ . Extensive simulation code, allowing randomized ground-truth cycle configurations, is published in Chung et al. (2026a): <https://github.com/laplcebeltrami/harmonic>.



**Fig. 4.** Top 10 cycles out of 534 identified from the mean harmonic flow  $\bar{U}$  across 400 human brain networks ( $p < 10^{-29}$ ). The cycles repeatedly involve shared brain regions but not identical directed edges, indicating reproducible cyclic organization at the node level rather than exact edgewise recurrence. This explains why direct edge-weight averaging fails to align cycles.



**Fig. 5.** Dense interacting-cycle simulation with three broken cycles. One edge per cycle is weakened below the noise level, masking cyclic structure in the observations. Existing methods, including transfer entropy, fail to recover the cyclic organization, whereas harmonic projection recovers the underlying cycles.



**Fig. 6.** Harmonic projection substantially improves cyclic recovery across baseline directional estimators and noise levels, demonstrating stable recovery of recurrent organization even when individual cycle edges are attenuated below the noise level.

## References

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